

Personal Information Manager

My Python Learning Project

Project Overview

This is a complete contact management system that I built to practice Python! It's like a mini-database where you can store information about people - their names, ages, cities, and hobbies. Everything gets saved to a file so your data is always there when you come back.

What I Learned Building This

1. File Handling

- Reading and writing JSON files
- Checking if files exist before loading
- Handling file errors gracefully

2. Data Structures

- Working with lists to store multiple records
- Using dictionaries for each person's information
- Managing IDs to keep track of records

3. Functions & Organization

- Breaking code into smaller, reusable functions
- Creating a class to organize related functions
- Building a menu system for user navigation

4. User Input & Validation

- Getting different types of input (text, numbers, lists)
- Validating age is a number and within range
- Preventing empty required fields

5. Search & Filter

- Finding records by name
- Case-insensitive searching
- Displaying filtered results

6. Data Analysis

- Calculating averages
- Counting unique values
- Finding most common items using Counter

How to Run

bash

Step 1: Make sure Python 3 is installed

python --version

Step 2: Navigate to project folder

cd path/to/project

Step 3: Run the program

python personal_info_manager.py

Step 4: Follow the menu options!

'''

Features Breakdown

Feature

Add Person - Create new records with validation

Display All - See all stored contacts at once

Search | Find people by name (partial match works!)

Update - Edit existing records, skip fields to keep old values

Delete - Remove records with safety confirmation

Statistics- View analytics: total count, average age, popular hobbies

Sample Session

'''

PERSONAL INFORMATION MANAGER

1. Add New Person

2. Display All Records

3. Search Person

4. Update Person
5. Delete Person
6. Show Statistics
7. Exit

=====

Enter your choice (1-7): 1

=====

ADD NEW PERSON

=====

Enter name: Alice Johnson

Enter age: 28

Enter city: Seattle

Enter hobbies (comma-separated): hiking, photography, cooking

- Alice Johnson added successfully!
- Data saved successfully!

Challenges I Faced & How I Solved Them

Challenge 1: How do I save data so it doesn't disappear when the program closes?

- **Solution:** Used JSON files! They're easy to read and write in Python with the json module

Challenge 2: What if someone enters "twenty" instead of "20" for age?

- **Solution:** Added try-except blocks to catch ValueError and show helpful error messages

Challenge 3: How do I let users update only some fields without re-entering everything?

- **Solution:** Added "leave blank to keep current value" logic - checks if input is empty before updating

Challenge 4: IDs kept getting messed up after deleting records

- **Solution:** Used len(self.people) + 1 for new IDs (Note: Could improve this in future versions!)

What I'd Improve Next

- Better ID management (reuse deleted IDs or use UUIDs)
- Export data to CSV format
- Add email and phone number fields

- Filter by city or age range
- Sort records by different criteria
- Add a graphical interface

Files Created

- `personal_info_manager.py` - Main program code
- `people_data.json` - Automatically created to store all records